

① ii) $p(s) = s^6 - s^5 - 6s^4 - s^2 - s + 6 = 0$

$$\begin{array}{l|rrrr} s^6 & 1 & -6 & -1 & 6 \\ s^5 & -1 & 0 & -1 & \\ s^4 & -6 & -2 & 6 & \\ s^3 & 1/3 & -2 & & \\ s^2 & -38 & & & \\ s^1 & -74/38 & & & \\ s^0 & 6 & & & \end{array}$$

No. of sign changes = 4

Total roots = 6

$\Rightarrow$ 2 roots in LHP 4
4 roots in RHP

ii) $p(s) = s(s^6 - s^5 - 4s^4 + 2s^3 + 7s^2 - 3s - 6)$

$$\begin{array}{l|rrrr} s^6 & 1 & -4 & 7 & -6 \\ s^5 & -1 & 2 & -3 & \\ s^4 & -2 & 4 & -6 & \\ s^3 & 0 & 0 & 0 & \\ s^3 & -1 & 1 & & \\ s^2 & 2 & -6 & & \\ s^1 & -2 & & & \\ s^0 & -6 & & & \end{array}$$

$$-2s^4 + 4s^2 - 6$$

$$\left(\frac{d}{ds} \right)$$

$$-4 + 12$$

$$-8s^3 + 8s - 0$$

$$= -s^3 + s$$

$\Rightarrow$ LHP = 3, RHP = 3, $x=0 \Rightarrow 1$

(iii) $p(s) = s^7 + 2s^6 + 5s^5 - 4s^4 + 8s^3 + 4s^2 - 6s$
 $= s(s^6 + 2s^5 + 5s^4 - 4s^3 + 8s^2 + 4s - 6)$

$$\begin{array}{l|rrrr} s^6 & 1 & 5 & 8 & -6 \\ s^5 & 2 & -1 & 4 & \\ s^4 & 7 & 6 & -6 & \\ s^3 & -40/7 & 40/7 & & \\ s^2 & 13 & -6 & & \\ s^1 & 40/13 & & & \\ s^0 & -6 & & & \end{array}$$

$x=0 \Rightarrow 1$, LHP = 3, RHP = 3

(iv) $P(s) = s^7 - 2s^6 - 5s^5 + 4s^4 + 8s^3 - 4s^2 - 6s$
 $= s(s^6 - 2s^5 - 5s^4 + 4s^3 + 8s^2 - 4s - 6)$

s^6	1	-5	+8	-6
s^5	-2	+4	-4	
s^4	-3	+6	-6	
s^3	-1	+1		
s^2	3	-6		
s^1	-1			
s^0	-6			

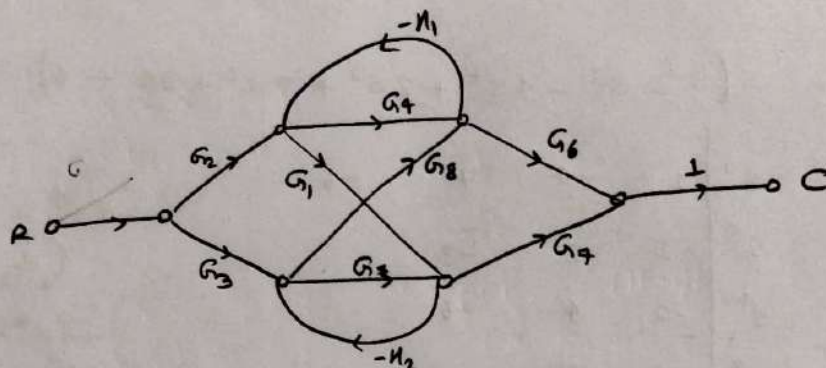
$$\frac{d}{ds}(-3s^4 + 6s^2 - 6) = -12s^3 + 12s$$

$$= -s^3 + s$$

3 RHP, 3 LHP, $1 \Rightarrow \gamma = 0$

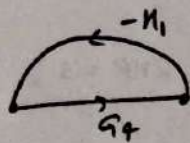
②

$$\frac{C(s)}{R(s)}$$

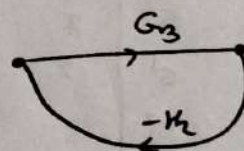


Loops $\Rightarrow$

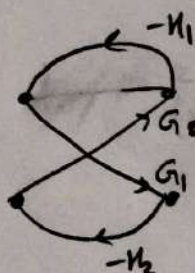
(i) L_1



(iii) L_2



(iii) L_3



$$\Delta = 1 - \sum L_i + \sum L_i L_j - \dots$$

$$\Delta = 1 - (-H_1 G_4 - H_2 G_3 + H_2 H_1 G_1 G_8)$$

$$+ H_1 H_2 G_3 G_4$$

$$\Rightarrow \Delta = 1 + H_1 G_4 + H_2 G_3 + H_1 H_2 G_3 G_4 - H_1 H_2 G_1 G_8$$

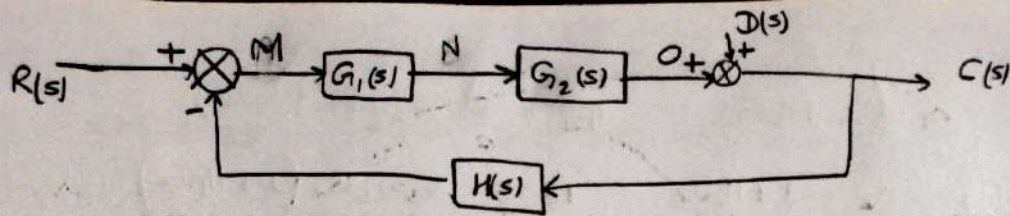
Forward paths :

$$\begin{aligned}
 \text{(i)} \quad P_1 &= G_2 G_4 G_6 & \Delta_1 &= 1 + H_2 G_5 & P_1 \Delta_1 &= G_2 G_4 G_6 (1 + H_2 G_5) \\
 \text{(ii)} \quad P_2 &= G_3 G_5 G_7 & \Delta_2 &= 1 + H_1 G_4 & P_2 \Delta_2 &= G_3 G_5 G_7 (1 + H_1 G_4) \\
 \text{(iii)} \quad P_3 &= G_1 G_2 G_7 & \Delta_3 &= 1 & P_3 \Delta_3 &= G_1 G_2 G_7 \\
 \text{(iv)} \quad P_4 &= G_3 G_6 G_8 & \Delta_4 &= 1 & P_4 \Delta_4 &= G_3 G_6 G_8 \\
 \text{(v)} \quad P_5 &= G_2 G_1 (-H_2) G_8 G_6 & \Delta_5 &= 1 & P_5 \Delta_5 &= -H_2 G_1 G_2 G_6 G_8 \\
 \text{(vi)} \quad P_6 &= G_3 G_8 (-H_1) G_1 G_7 & \Delta_6 &= 1 & P_6 \Delta_6 &= -H_1 G_1 G_3 G_7 G_8
 \end{aligned}$$

Transfer function $\Rightarrow \frac{\sum P_k \Delta_k}{\Delta}$

$$G(s) = \frac{
 \begin{aligned}
 &G_2 G_4 G_6 (1 + H_2 G_5) + G_3 G_5 G_7 (1 + H_1 G_4) \\
 &+ G_1 G_2 G_7 + G_3 G_6 G_8 + H_2 G_1 G_2 G_6 G_8 - H_1 G_1 G_3 G_7 G_8
 \end{aligned}
 }{
 \begin{aligned}
 &(1 + H_1 G_4 (1 + H_2 G_5) + H_2 G_5 - H_1 H_2 G_1 G_8)
 \end{aligned}
 }$$

③



$$E(s) = R(s) - C(s)$$

$$e(\infty), R(s) = D(s) = U(s)$$

$$M(s) = R(s) - CH \Rightarrow M = R - CH$$

$$N = G_1(M) = G_1(R - CH)$$

$$O = G_2 G_1 (R - CH)$$

$$C = O + D = G_2 G_1 (R - CH) + D$$

$$\Rightarrow C = \frac{G_1 G_2 R + D}{1 + G_1 G_2 H}$$

p)

$$E(s) = R(s) - C(s)$$

$$= R - \left(\frac{G_1 G_2 R + D}{1 + G_1 G_2 H} \right) = \frac{R(1 + G_1 G_2 H) - D}{1 + G_1 G_2 H}$$

b)

$$R(s) = D(s) = \frac{1}{s}$$

$$e(\infty) = \lim_{s \rightarrow 0} s E(s) = s \times \left(\frac{\frac{1}{s} (1 + G_1 G_2 H) - \frac{1}{s}}{1 + G_1 G_2 H} \right)$$

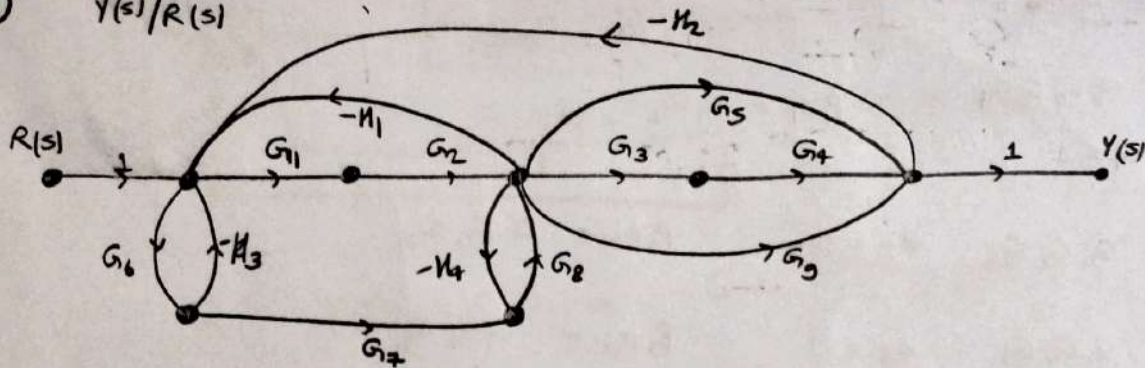
$$= \frac{G_1 G_2 H - 1}{1 + G_1 G_2 H} \Rightarrow \frac{G_1(0) G_2(0) (H(0) - 1)}{1 + G_1(0) G_2(0) H(0)}$$

c)

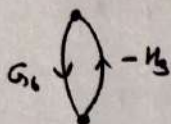
$$e(\infty) = 0 \Rightarrow H(0) = 1 \quad \& \quad G_1(0) G_2(0) H(0) \neq -1$$

⑤

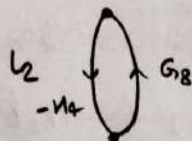
$Y(s)/R(s)$



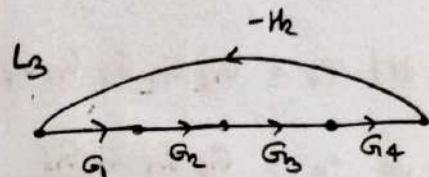
(i) L_1



(ii)

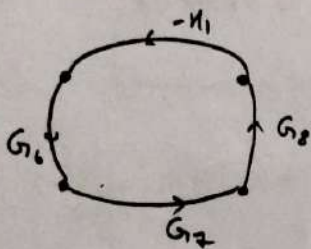


(iii)

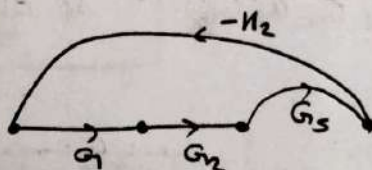


(iv)

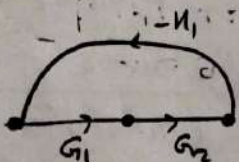
L_4



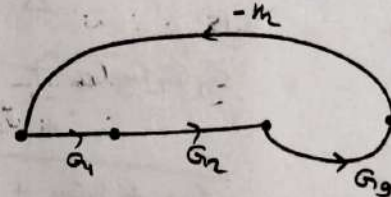
(v) L_5



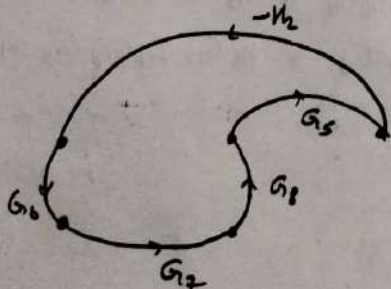
(vi) L_6



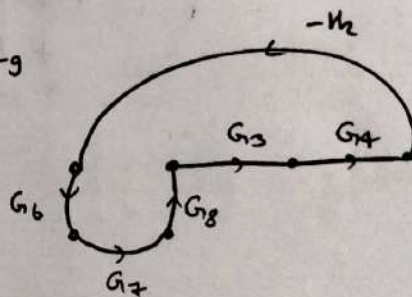
(vii) L_7



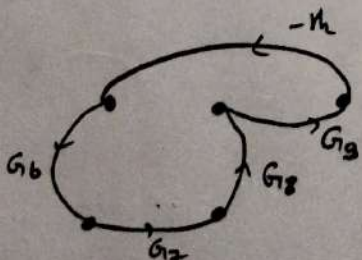
(viii) L_8



(ix) L_9



(x)



$$\Delta = 1 - \sum L_i + \sum L_i L_j - \sum L_i L_j L_k \dots$$

$$\Delta = 1 - (-H_3 G_6 - H_4 G_8 - H_2 G_1 G_2 G_3 G_4 - H_1 G_6 G_7 G_8 - H_2 G_1 G_2 G_3 G_4 G_5 - H_1 G_1 G_2 - H_2 G_1 G_2 G_3 G_4 G_5 - H_2 G_6 G_7 G_8 G_9 - H_2 G_3 G_4 G_5 G_6 G_7 G_8 - H_2 G_6 G_7 G_8 G_9) + H_3 H_4 G_6 G_8$$

Forward paths

$$\text{i)} P_1 = G_1 G_2 G_3 G_4 \\ \Delta_1 = 1$$

$$P_1 \Delta_1 = G_1 G_2 G_3 G_4$$

$$\text{ii)} P_2 = G_1 G_2 G_5 \quad \Delta_2 = 1$$

$$P_2 \Delta_2 = G_1 G_2 G_5$$

$$\text{iii)} P_3 = G_1 G_2 G_9 \quad \Delta_3 = 1$$

$$P_3 \Delta_3 = G_1 G_2 G_9$$

$$\text{iv)} P_4 = G_6 G_7 G_8 G_3 G_4 \quad \Delta_4 = 1$$

$$P_4 \Delta_4 = G_6 G_7 G_8 G_3 G_4$$

$$\text{v)} P_5 = G_6 G_7 G_8 G_5 \quad \Delta_5 = 1$$

$$P_5 \Delta_5 = G_6 G_7 G_8 G_5$$

$$\text{vi)} P_6 = G_6 G_7 G_8 G_9 \quad \Delta_6 = 1$$

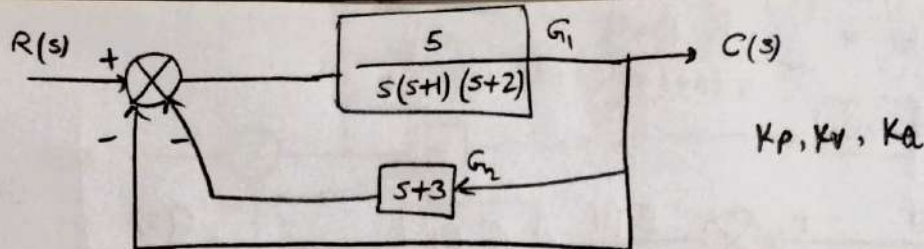
$$P_6 \Delta_6 = G_6 G_7 G_8 G_9$$

$$\text{Transfer function} = \frac{\sum P_k \Delta_k}{\Delta}$$

$$= \frac{G_1 G_2 G_3 G_4 + G_1 G_2 G_5 + G_1 G_2 G_9 + G_6 G_7 G_8 G_3 G_4 + G_6 G_7 G_8 G_5 + G_6 G_7 G_8 G_9}{\Delta}$$

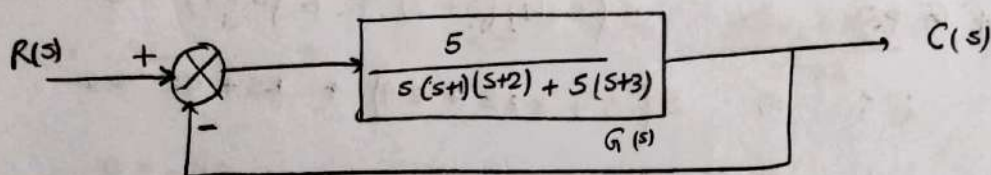
$$\Delta = 1 + H_3 G_6 + H_4 G_8 + H_2 G_1 G_2 G_3 G_4 + H_1 G_6 G_7 G_8 + H_2 G_1 G_2 G_5 + H_1 G_1 G_2 + H_2 G_1 G_2 G_9 + H_2 G_6 G_7 G_8 G_5 + H_2 G_3 G_4 G_8 G_7 G_8 + H_2 G_6 G_7 G_8 G_9 + H_3 H_4 G_6 G_9$$

⑥



$$G_1(R - G_2 C) = C \Rightarrow C = \frac{G_1 R}{1 + G_1 G_2}$$

$$\Rightarrow C = \frac{5}{s(s+1)(s+2) + 5(s+3)}$$



$$(a) K_p = \lim_{s \rightarrow 0} G(s) = \lim_{s \rightarrow 0} \frac{5}{s(s+1)(s+2) + 5(s+3)} = \frac{5}{5 \times 3} = \frac{1}{3} = K_p$$

$$K_v = \lim_{s \rightarrow 0} s G(s) = \lim_{s \rightarrow 0} \frac{5s}{s(s+1)(s+2) + 5(s+3)} = \frac{5(0)}{2 + 15} = 0$$

$$K_a = \lim_{s \rightarrow 0} s^2 G(s) = \lim_{s \rightarrow 0} \frac{5s^2}{s(s+1)(s+2) + 5(s+3)} = 0$$

$$(b) e_{ss} = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} \frac{s U(s)}{1 + G(s)} = \lim_{s \rightarrow 0} \frac{1 |U(s)|}{1 + G(s)}$$

$$= \lim_{s \rightarrow 0} \frac{1 |U(s)|}{1 + \frac{5}{0 + 15}} = \lim_{s \rightarrow 0} \frac{|U(s)|}{1 + \frac{1}{3}} = \frac{3}{4} |U(s)|$$

$$50 u(t) \Rightarrow \frac{3}{4} \times 50 = \frac{75}{2} = e_{ss}$$

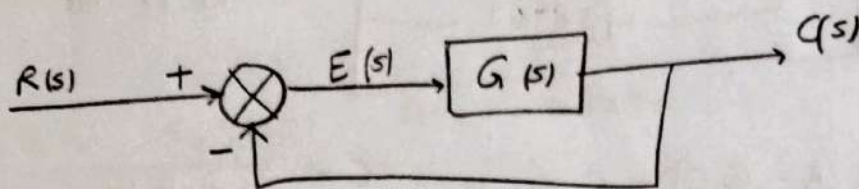
$$50t u(t) \Rightarrow \frac{\frac{50}{s^2} \times s}{1 + G(s)} \Rightarrow \frac{50}{s(1 + G(s))} = \frac{50}{0 + K_v} = \infty = e_{ss}$$

$$50t^2 u(t) \Rightarrow \frac{\frac{50 \cdot 2 \cdot s}{s^3}}{1 + G(s)} \Rightarrow \frac{50}{K_a} = \infty = e_{ss}$$

(c) The system is type 0.

7

$$G(s) = \frac{K(s+4)}{s(s+1.2)(s+2)}$$



$$\frac{C(s)}{R(s)} = \frac{G}{1+G}$$

$$1+G=0 \Rightarrow s(s+1.2)(s+2) + K(s+4) = 0 \text{ , roots}$$

$$\Rightarrow s(s^2 + 3.2s + 2.4) + Ks + 4K = 0$$

$$\Rightarrow s^3 + 3.2s^2 + (K+2.4)s + 4K = 0$$

$$\Rightarrow 5s^3 + 16s^2 + (5K+12)s + 20K = 0$$

Routh Table (RT)

s^3	5	$5K+12$
s^2	16	$20K$
s^1	$\frac{-100K+80K+192}{16} = \frac{192-20K}{16}$	
s^0	$20K$	

Roots in LHP $\Rightarrow$ no sign change $\Rightarrow \frac{192-20K}{16} > 0$

$$192 > 20K \Rightarrow K < 9.6$$

(a)

Stable system $0 < K < 9.6$

(b)

To make the system oscillate $\Rightarrow \frac{192-20K}{16} = 0 \Rightarrow K = 9.6$

(c)

$$T(s) = \frac{G}{1+G} = \frac{K(s+4)}{s(s+1.2)(s+2) + K(s+4)} = \frac{48(s+4)}{5s^3 + 16s^2 + 60s + 192}$$

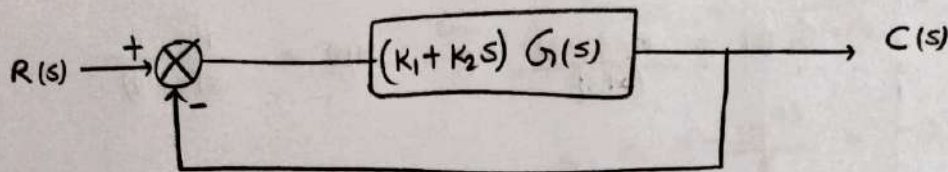
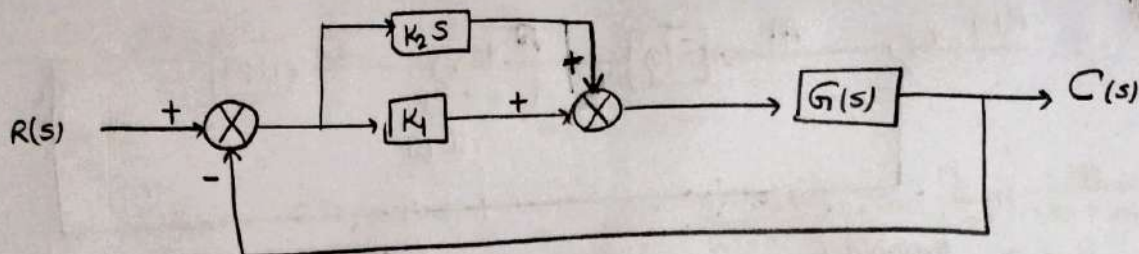
doing $16s^2 + 20K = 0$

$$\Rightarrow 16(j\omega)^2 + 20 \times 9.6 = 0$$

$$\Rightarrow -16\omega^2 = -192$$

$$\Rightarrow \omega^2 = 12 \Rightarrow \omega = \sqrt{12}$$

9



$$G(s) = \frac{2}{s^2 + 2} \Rightarrow (K_1 + K_2 s) G(s) = \frac{2(K_1 + K_2 s)}{s^2 + 2}$$

(i) Stable System

$$G(s) = \frac{2(K_1 + K_2 s)}{s^2 + 2} \Rightarrow T(s) = \frac{2(K_1 + K_2 s)}{s^2 + 2 + 2(K_1 + K_2 s)}$$

$$\Rightarrow T(s) = \frac{2(K_1 + K_2 s)}{s^2 + 2K_2 s + 2(1 + K_1)} \longrightarrow \text{Roots in LHP}$$

$$\begin{array}{c|c} s^2 & 1 \\ s^1 & 2K_2 \\ s^0 & 2 + 2K_1 \end{array}$$

$$\Rightarrow 2K_2 > 0 \Rightarrow K_2 > 0$$

$$\Rightarrow 2 + 2K_1 > 0 \Rightarrow K_1 > -1$$

↑
stable system

(ii) $e_{ss} \leq \frac{1}{100}$ $\uparrow$ $\gamma = \frac{1}{\sqrt{10}}$

$$e_{ss} = \frac{1}{1 + K_p} \quad \text{where} \quad K_p = \lim_{s \rightarrow 0} (K_1 + K_2 s) G(s)$$

$$= \lim_{s \rightarrow 0} \frac{(K_1 + K_2 s) 2}{s^2 + 2} = \frac{2K_1}{2} = K_1$$

$$\Rightarrow \frac{1}{1 + K_1} \leq \frac{1}{100}$$

$$\Rightarrow K_1 \geq 99$$

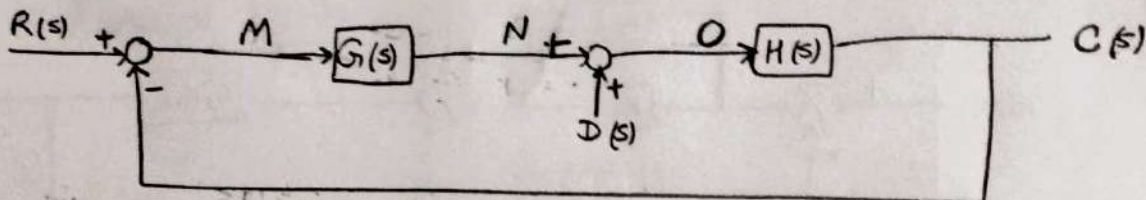
$$\gamma = \frac{1}{\sqrt{10}} \Rightarrow \zeta K_2 = \zeta \omega_n \Rightarrow K_2 = \frac{\zeta \omega_n}{\omega_n} = \frac{\zeta K_1}{2} = K_1$$

$$\Rightarrow K_2^2 = \frac{1}{10} 2(1 + K_1) = \frac{1 + K_1}{5}$$

$$\Rightarrow \text{As } K_1 \geq 99 \Rightarrow K_2^2 \geq \frac{100}{5} \Rightarrow K_2 \geq \sqrt{20}$$

$$K_1 \geq 99$$

10.



$$e_{ss} = 0.000012$$

$U(t)$

$$e_{ss} = 0.003$$

(ramp)

$$G(s) = \frac{K_1 (s+2)}{(s+3)}$$

$$H(s) = \frac{K_2}{s(s+4)}$$

$$M = R - C$$

$$N = G(R - C)$$

$$O = D + G(R - C)$$

$$C = HD + GH(R - C)$$

$$\Rightarrow C = \frac{HD + HGR}{1 + GH}$$

$$E(s) = -(C - R) = R - \frac{HD + HGR}{1 + GH} = \frac{R - HD}{1 + GH}$$

$$e_{ss} = \lim_{s \rightarrow 0} sE(s) = \cancel{s \left(\frac{1}{s} - \frac{K_2}{s(s+4)} \right)} = \frac{s+4-K_2}{(s+4)} = \frac{s+4-K_2}{s(s+3)(s+4) + K_1 K_2 (s+2)}$$

$$= \frac{s(s+4-K_2)(s+3)}{s(s+3)(s+4) + K_1 K_2 (s+2)} = 0.000012$$

$$E_R(s) = \frac{R(s)}{1 + GH}$$

$$E_D = \frac{-DH}{1 + GH}$$

$$(i) e_{ss} = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} \frac{s \times -1/s \times \frac{K_2}{s(s+4)}}{1 + \frac{K_1(s+2)K_2}{s(s+3)(s+4)}}$$

$$D = 1/s$$

$$= \lim_{s \rightarrow 0} \frac{-K_2/s(s+4)}{1 + \frac{2K_1K_2}{12s}} = \lim_{s \rightarrow 0} \frac{-K_2/4s}{\frac{12s + 2K_1K_2}{12s}} = \lim_{s \rightarrow 0} \frac{-3K_2}{12s + 2K_1K_2}$$

$$= \lim_{s \rightarrow 0} \frac{-3K_2}{12s + 2K_1K_2} = \frac{-3K_2}{2K_1K_2} = 0.000012$$

$$\Rightarrow K_1 = -\frac{10^6}{8} = -125,000$$

$$(ii) e_{ss} = 0.003$$

$$e_{ss} = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} \frac{s R(s)}{1 + GH} \quad R(s) = \frac{1}{s^2}$$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s \times 1/s^2}{1 + \frac{K_1K_2(s+2)}{s(s+4)(s+3)}} = \frac{(s+4)(s+3)}{s(s+4)(s+3) + K_1K_2(s+2)}$$

$$= \frac{12 \cdot 6}{2K_1K_2} = 0.003$$

$$\Rightarrow K_2 = \frac{6}{0.003 K_1} = \frac{6}{(0.003)(-125,000)} = -16 \times 10^{-3}$$

$$\Rightarrow K_1 = -1.25 \times 10^5 \quad K_2 = -1.6 \times 10^{-2}$$

③ ii) $G_1(s) = \frac{K}{s(s+10)(s^2 + 20s + 500)}$

$$p(s) = s(s+10)(s^2 + 20s + 500) + K$$

$$= s^4 + 30s^3 + 700s^2 + 5000s + K$$

s^4	1	700	K
s^3	30	5000	
s^2	$\frac{1600}{3}$	K	
s	$30K - \frac{8 \times 10^6}{3}$		
		$-\frac{1600}{3}$	
1		K	

No sign changes,

$$\Rightarrow K > 0$$

4

$$\frac{8 \times 10^6}{3} - 30K > 0$$

$$\Rightarrow K < \frac{8}{9} \times 10^5$$

$$\Rightarrow 0 < K < \frac{8}{9} \times 10^5$$

$$(ii) \quad (s^4 + 30s^3 + 700s^2 + 5000s)(s^2 + 80s + 2000) + K(s^2 + 40s + 600)$$

$$s^6 + 110s^5 + s^4(2000 + 2400 + 700) + s^3(121000)$$

$$+ s^2(1800000) + 10000000s + K(s^2 + 40s + 600)$$

s^6	1	5100	$1800000 + K$	$600K$
s^5	110	121000	$10000000 + 40K$	0
s^4	4000	$\frac{18800000 + 7K}{11}$	$600K$	0
s^3	$74000 - \frac{7K}{400}$	$10,000,000 + 23.5K$	0	0
s^2	$\frac{-\frac{1}{11} \left\{ 4K^2 + 338000000K - 3.8048 \times 10^9 \right\}}{29600000 - 7K}$		$600K$	
s^1	$40 \left[\frac{-9.512 \times 10^{13} + 2.22384 \times 10^{14}K + 3.9885 \times 10^7 K^2 + 49K^2}{49K^2 + 3.38 \times 10^8 K - 3.8048 \times 10^{14}} \right]$			
s^0	600K			

setting first column $> 0 \Rightarrow K < 3.88 \times 10^6$

Hence the available range of values of K is very high!

④

$$P(s) = s^n + a_1 s^{n-1} + a_2 s^{n-2} + \dots + a_{n-1} s + a_n = 0$$

Routh Table

$$\begin{array}{c|ccccccccc} s^n & 1 & a_2 & a_4 & a_6 & a_8 & \dots \\ s^{n-1} & a_1 & a_3 & a_5 & & a_7 & a_9 & \dots \\ s^{n-2} & b_1 & b_2 & b_3 & & b_4 & b_5 & \dots \\ s^{n-3} & c_1 & c_2 & c_3 & & c_4 & c_5 & \dots \\ & 1 & & & & & & \\ & 1 & & & & & & \\ & 1 & a_n & & & & & \end{array}$$

$$b_1 = \frac{\begin{vmatrix} 1 & a_2 \\ a_1 & a_3 \end{vmatrix}}{-a_1}$$

$$b_2 = \frac{\begin{vmatrix} 1 & a_4 \\ a_1 & a_5 \end{vmatrix}}{-a_1}$$

$$Q = \frac{\begin{vmatrix} a_1 & a_3 \\ b_1 & b_2 \end{vmatrix}}{-b_1}$$

$$C_2 = \frac{\begin{vmatrix} a_1 & a_5 \\ b_1 & b_3 \end{vmatrix}}{-b_1}$$

$$\Delta_1 = a_1 \quad b_1 = \frac{\begin{vmatrix} a_1 & a_3 \\ 1 & a_2 \end{vmatrix}}{a_1} = \frac{\Delta_2}{\Delta_1}$$

$$\Delta_2 = \begin{vmatrix} a_1 & 1 \\ a_3 & a_2 \end{vmatrix}$$

$$G = \frac{\begin{vmatrix} b_1 & a_1 \\ b_2 & a_3 \end{vmatrix}}{b_1}$$

$$= \frac{\begin{vmatrix} \frac{1}{a_1} & a_1 & 1 \\ \frac{1}{a_1} & a_3 & a_2 \end{vmatrix}}{b_1} \quad \begin{vmatrix} a_1 \\ a_3 \end{vmatrix}$$

$$= \frac{(a_1 a_2 - a_3) a_3}{a_1} - (a_1 a_4 - a_5) / b_1$$

$$= \frac{(a_1 a_2 - a_3) a_3 - a_1 (a_1 a_4 - a_5)}{b_1/a_1} \quad \Delta_2$$

$$\Delta_3 = \begin{vmatrix} a_1 & 1 & 0 \\ a_3 & a_2 & a_1 \\ a_5 & a_4 & a_3 \end{vmatrix} = a_1(a_2 a_3 - a_1 a_4) - (a_3^2 - a_1 a_5)$$

$\Rightarrow Q = \frac{\Delta_2}{\Delta_1}$, similarly it can be shown for all the entries of the 1st column.